DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING(A)
Osmania University, Hyderabad-07
CLASS TEST-2

SUBJECT: Automata Languages and Computation
Class: B.E(CSE)
Semester: IV

Date: 05-05-2026
Max. Marks: 30
Duration: 1.5 Hours

S.No.	PART-A Answer All Questions(5x2=10M)	Marks	BT	CO
1. a)	Define Grammar. Write the production rules of CNF, GNF.	2	1	2
b)	What is Unit Production? Write example of eliminating Unit Production.	2	2	2
c)	Define LBA with neat diagram.	2	1	1
d)	Write differences between PDA and TM.	2	4	3
e)	Write the definition of Type-0 grammar.	2	2	2
	PART-B **Answer any Two Questions(10X2=20M)**			
2.	a)Give rules for Pumping Lemma on CFL. Prove the language L={www/wϵ(a,b)*} is not a context free language.	5	3	2
	b)Convert the given grammar into Greibach Normal Form (GNF). S→AA|B, A→ aA|B | εB→a | ε	5	4	2
3.	a)DesignTM for L = $\{a^n b^m c^m d^n, n, m >= 1\}$. Construct ID for sample string.	6	6	1
	b)Describe Modifications or Types of Turing machine.	4	2	3
4.	a)Explain Chomsky hierarchy of languages with neat diagram.	5	2	2
	b)Give instance of PCP for the given problem. Identify it is having a solution or not. A= {11,100,111}, B= {111,001, 11}	5	5	4

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